

Beyond Basic - Part 5

Written by Administrator

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This issue I will start to detail the collision detection routine to go with the automatic sprite routines from last issue. But first here is a list of the conversions for SVI-318/328 to MSX.

Part 1

The SVI-318/328 use the port 8C to change slots. So to change between the RAM and ROM slots use:

ROM to RAM	RAM to ROM
LD A,0FH	LD A,(FE64H)
OUT (88H),A	OUT (8CH),A
IN A,(90H)	
LD (FE64H),A	
AND FDH	
OUT (8CH),A	

Part 2 Onwards

Here is a list of ROM calls, RAM locations and HOOK addresses to use instead of the MSX values.

- HGONE = FF57H,
- HTIMI = FF5AH,
- SCRMOD = FE3AH
- SETWRT = 373CH,
- SETRD = 3747H

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Change the following port calls to:

- OUT (98H),A becomes OUT (80H),A
- IN A,(98H) becomes IN A,(84H)

This list will be expanded if necessary, each issue.

Sprite Collision Detection

Now onto this months routine. Our routine will test the collision of one sprite (you specify) with all 31 others. A table of flags will be used to show which sprites the one specified is colliding with. The size of each sprite will be specified by a X and Y with from 0 to 15 (There is no zero width, work out the real value and subtract one). For the routine to work effectively, your sprite shapes should be drawn from the top left hand corner of the sprite shape pattern.

e.g. X width 5, Y width 2

To make the routine easier to understand I have split it into two sections. The section I will show this issue is the actual test routine, that tests whether two specified sprites are colliding and setting the carry flag if so. The values of the registers on entry are:

- HL points to sprite 1 VRAM table location
- DE points to sprite 2 VRAM table location
- B lower 4 bits, y width and top 4 bits, x width for sprite 1
- C lower 4 bits, y width and top 4 bits, x width for sprite 2

Machine Code Program

1 ;	Subroutine to test the collision of two sprites		
2 ;			
3 ;			
4 ;			
5 ; Label settings			
6 ;			
7 SETRD	EQU 0050H	; Video read	
8 ;			
C0FA	E5	9 START	PUSH HL ; Save registers

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C0FB	D5	10	PUSH DE	;
C0FC	C5	11	PUSH BC	;
C0FD	78	12	LD A,B	; Get Y velocity 1
C0FE	E60F	13	AND 15	;
C100	47	14	LD B,A	;
C101	79	15	LD A,C	; Get Y velocity 2
C102	E60F	16	AND 15	;
C104	4F	17	LD C,A	;
C105	CD55C1	18	CALL RDVRM	; Get Y1 value
C108	80	19	ADD A,B	; Add velocity
C109	47	20	LD B,A	;
C10A	EB	21	EX DE,HL	; Get Y2 value
C10B	CD55C1	22	CALL RDVRM	;
C10E	EB	23	EX DE,HL	;
C10F	B8	24	CP B	; Hit?
C110	3040	25	JR NC,NOHIT	; No – Exit loop
C112	EB	26	EX DE,HL	; Get Y2 value
C113	CD55C1	27	CALL RDVRM	;
C116	EB	28	EX DE,HL	;
C117	81	29	ADD A,C	; Add velocity
C118	4F	30	LD C,A	;
C119	CD55C1	31	CALL RDVRM	; Get Y1 velocity
C11C	B9	32	CP C	; Hit?
C11D	3033	33	JR NC,NOHIT	; No – Exit loop
C11F	C1	34	POP BC	; Restore BC
C120	C5	35	PUSH BC	; Resave BC
C121	23	36	INC HL	; Increment pointers
C122	13	37	INC DE	;
C123	CB38	38	SRL B	; Get X velocity 1
C125	CB38	39	SRL B	;
C127	CB38	40	SRL B	;
C129	CB38	41	SRL B	;
C12B	CB39	42	SRL C	; Get X velocity 2
C12D	CB39	43	SRL C	;
C12F	CB39	44	SRL C	;
C131	CB39	45	SRL C	;
C133	CD55C1	46	CALL RDVRM	; Get Y1 value
C136	80	47	ADD A,B	; Add velocity
C137	47	48	LD B,A	;
C138	EB	49	EX DE,HL	; Get Y2 velocity
C139	CD55C1	50	CALL RDVRM	;
C13C	EB	51	EX DE,HL	;
C13D	B8	52	CP B	; Hit?
C13E	3012	53	JR NC,NOHIT	; No – Exit loop
C140	EB	54	EX DE,HL	; Get Y2 value
C141	CD55C1	55	CALL RDVRM	;

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C144	EB	56	EX DE,HL	;
C145	81	57	ADD A,C	; Add velocity
C146	4F	58	LD C,A	;
C147	CD55C1	59	CALL RDVRM	; Get Y1 value
C14A	B9	60	CP C	; Hit?
C14B	3005	61	JR NC,NOHIT	; No – Exit loop
C14D	37	62	SCF	; Set carry flag
C14E	C1	63 EXIT	POP BC	; Restore registers
C14F	D1	64	POP DE	;
C150	E1	65	POP HL	;
C151	C9	66	RET	; Return from routine
C152	AF	67 NOHIT	XOR A	; Clear flags
C153	18F9	68	JR EXIT	; Go to EXIT
C155	69 ;			
C155	70 ;		Subroutine to read a byte from VRAM	
C155	71 ;			
C155	CD5000	72 RDVRM	CALL SETRD	; Set screen location
C158	D898	73	IN A,(98H)	; Get value
C15A	C9	74	RET	;
C15B	75 ;			
	76 END			